Haoran Du

Phone: +1 (343) 368 2458
Email: du_haoran@outlook.com
Address: London, ON, CA
Eligiblity: Chinese Citizenship, Canadian Permanent Residency
ORCID: 0000-0002-0848-7875

Education

Western University London, ON, Canada
Doctor of Philosophy in Mechanical Engineering May 2025 – Expected 2028
Academic standing: 95.75/100
Master of Engineering Science in Mechanical Engineering Sept. 2021 – Aug. 2023
Academic standing: 94.25/100

University of Pittsburgh Pittsburgh, PA, USA
Master of Data Science in Machine Learning May 2024 – Expected 2026
Academic standing: 4.00/4.00

University of Cincinnati Cincinnati, OH, USA
Bachelor of Science in Mechanical Engineering Sept. 2015 – May 2020
Academic standing: 3.93/4.00

Chongqing University Chongqing, China
Bachelor of Engineering in Mechanical Engineering Sept. 2015 – Jun. 2020
Academic standing: 3.83/4.00

Coursework

Engineering Science: Acoustics, Applied Computational Fluid Dynamics and Heat Transfer, Computational Wind Engineering, Convection Heat Transfer, Experimentation & Measurement, Fluid Mechanics, Wind Engineering, Gas Turbine Combustion, Internal Combustion Engine, Modeling & Simulation of Multi-Physics Systems, Viscous Boundary Layer Flow
Data Science: Data-Centric Computing, Responsible Data Science, Mathematical & Statistical Foundations for Data Science, Art of Data Visualization, Managing, Querying & Preserving Data, Applied Predictive Modeling, Applied Bayesian Data Analysis, Applied Deep Learning, Large Language Models and Their Applications, Machine Learning for Engineering Applications

Research Experience

- **Western University** London, ON, CA
  Graduate Research Assistant (full-time) May 2025 – Present
  - Developing Large-Eddy Simulations (LES) for urban canopy flow, including idealized street canyons, tall buildings, with focus on turbulence, particulate pollutant dispersion, and convective heat transfer.
  - Developing Reduced-Order Models (ROMs), i.e. Autoencoder (AE), Convolutional Neural Network (CNN), etc. for urban canopy flow and pollutant dispersion through sensor-informed neural networks.

  Voluntary Research Assistant (part-time) Jan. 2024 – Apr. 2025
  - Developed LES simulations for atmospheric boundary layer flow using the drag porosity method.
  - Built predictive linear models of large-scale flow in atmospheric boundary layer in urban area through Proper Orthogonal Decomposition (POD) and Linear Stochastic Estimation (LSE).

  Graduate Research Assistant (full-time) Sep. 2021 – Aug. 2023
  - Analyzed the data from wind tunnel experiments to study the effect of a single upstream tall building on the mean turbulence statistics and scale interactions of the downstream boundary layer and street canyon.
  - Designed an experimental cough simulator and tested different actuation speeds using Particle Image Velocimetry (PIV) and Hot Wire Anemometry (HWA) technology in a cough chamber.
- **École Centrale de Nantes** Nantes, PDL, FR
  Visiting Research Assistant (full-time) Jun. 2026 – Aug. 2026
  - Designed and conducted experiments using PIV and HWA technology in a wind tunnel over an idealized model to study the street canyon flow.

  Visiting Research Assistant (full-time) Jun. 2023 – Aug. 2023
  - Analyzed the data from wind tunnel experiments to study the effect of a single tall building on the scale interactions between the downstream boundary layer and the street canyon flow.

Visiting Research Assistant (full-time) May 2022 – Aug. 2022

- Conducted experiments using PIV and HWA technology in a wind tunnel over a heterogeneous model (RdS) to study the street canyon flow.

- **University of Cincinnati** Cincinnati, OH, US
  Undergraduate Research Assistant (part-time) Sep. 2019 – Apr. 2020
  - Designed the mounting structure between a vertical-axis wind turbine and an automobile using SIEMENS-NX.
  - Assisted with Unsteady Reynolds-Averaged Navier-Stokes (URANS) simulations to examine the fluid field of the designed model in Star-CCM+.
- **Université Laval** Québec, QC, CA
  Undergraduate Research Assistant (co-op) May 2019 – Aug. 2019
  - Designed a connector between the aluminum extrusion deck and the timber girder in SolidWorks at the REGAL Aluminum Research Center.
  - Simulated Finite Element Analysis (FEA) of the connector stress under conditions of large temperature differences and live traffic load in Abaqus.
- **Chongqing University** Chongqing, CQ, CN
  Undergraduate Research Assistant (co-op) Jan. 2018 – Apr. 2019
  - Ran FEA in Abaqus of a gear pair and calculated fatigue life in Fe-safe.
  - Compared the performance parameters of two transmission systems built in Masta and proposed a planetary system with excellent performance.

Teaching Experience

- **Western University** London, ON, CA
  Graduate Teaching Assistant (part-time) Jan. 2022 – Present
  - Tutored for Intro to Fluid Mechanics & Heat Transfer (215 hours), Thermodynamics I (105 hours), Heat Tansfer II (70 hours), Fluid Machinery (70 hours), Product Design (55 hours), and System Modeling & Control (70 hours),
- **University of Cincinnati** Cincinnati, OH, US
  Undergraduate Teaching Assistant (part-time) Jan. 2020 – May 2020
  - Graded and tutored for Engineering Design Thinking II (100 hours).
- **Chongqing University** Chongqing, CQ, CN
  Undergraduate Teaching Assistant (co-op) Jan. 2018 – Apr. 2018
  - Graded and tutored for Statics and Dynamics (140 hours), Numerical Techniques (70 hours), Systematic Dynamics & Vibration (140 hours), and Heat Transfer (70 hours).

Work Experience

- **Stäubli** Hangzhou, ZJ, CN
  Application Engineer Assistant Dec. 2020 – Jun. 2021
  - Used Inventor for modeling and producing drawings.
  - Assembled and tested samples in the laboratory.
  - Assisted with project management.
- **Eccalc** Hangzhou, ZJ, CN
  Application Engineer Assistant (co-op) May 2017 – Aug. 2017
  - Cloudized online calculation for sheet metal engineering components.
- **TCL Rechi** Huizhou, GD, CN
  Mechanical Engineer Assistant (co-op) Sep. 2016 – Dec. 2016
  - Utilized AutoCAD for modeling and producing drawings.
  - Tested compressor models for air-conditioning in the laboratory.

Skills

CAD Software: SolidWorks, Siemens NX, Autodesk Inventor
Simulation Software: OpenFOAM, Ansys Fluent, Simcenter STAR-CCM+, Abaqus
Programming: Matlab, Python, R, MySQL
Languages: Chinese (Native), English (Advanced), French (Intermediate)

Awards

Canada Graduate Research Scholarship – Doctoral	2026 – present
Western Graduate Research Scholarship and Fellowship	2021–2023, 2025–present
Mitacs Globalink Graduate Fellowship	2021 – 2022
University of Cincinnati Summa Cum Laude	2020
Mitacs Globalink Research Internship Award	2019
CSC Visiting Student Award	2019
Chongqing University Comprehensive Excellence Scholarship	2017

Publications

Journal:

- **Du, H.**, Savory, E., Perret, L. (2023). Effect of morphology and an upstream tall building on the mean turbulence statistics of a street canyon flow. *Building and Environment.*
- **Du, H.**, Perret, L., Savory, E. (2024). Effect of urban morphology and an upstream tall building on the scale interaction between the overlying boundary layer and a street canyon. *Boundary-Layer Meteorology.*
- **Du, H.**, Perret, L., Savory, E. (2025). Investigation of boundary layer scale interactions over realistic morphology influenced by an upstream tall building using the SPOD method. *Boundary-Layer Meteorology.*
- **Du, H.**, Savory, E., Perret, L. (in peer-review). Data-driven reconstruction of large scales in atmospheric boundary layer flow with sparse measurements. *Experiments in Fluids*

Conference:

- **Du, H.**, Savory, E., Perret, L. (2022, August 29–31). Effect of an upstream tall building on a street canyon flow [Paper presentation]. PHYSMOD2022 – International Workshop on Physical Modeling of Flow and Dispersion Phenomena, Prague, Czechia.
- **Du, H.**, Perret, L., Savory, E. (2024, August 28–30). Scale interaction between the urban boundary layer and a street canyon in a morphological model [Paper presentation]. PHYSMOD2024 – International Workshop on Physical Modeling of Flow and Dispersion Phenomena, Lyon, France.
- **Du, H.**, Savory, E., Perret, L. (2025, May 25–28). Characteristics of street canyon flow using Large-Eddy Simulation with a drag-porosity modelled atmospheric boundary layer [Paper presentation]. CFDSC2025 – 32nd Annual Conference of the CFD Society of Canada, Montréal, Canada.
- **Du, H.**, Savory, E., Perret, L. (2026, August 25–28). Reconstruction of large scales in atmospheric boundary layer flow with sparse measurements [Paper presentation]. PHYSMOD2026 –International Workshop on Physical Modeling of Flow and Dispersion Phenomena, Potsdam, Germany.

Presentations

Symposium:

- **Du, H.** (2026, May 4–5). Data-driven reconstruction of atmospheric boundary layer flow from sparse measurements [Oral presentation]. 2026 Annual Graduate Student Symposium, Department of Mechanical and Materials Engineering, Western University, London, Ontario, Canada.